TwinPulse: system architecture

Digital twin that rehearses tonight to prevent night-time hypoglycaemia in children with type 1 diabetes

1 DATA

Static: health record

- FHIR R4 bundle (ABDM-ready)
- Age, weight, HbA1c, Hb
- Insulin regimen, daily dose
- Severe lows, hypo awareness
- HLA genotype, autoantibodies

Dynamic: wearables

- CGM (5 min) or finger-pricks
- Insulin doses (pen / pump)
- Meals (carbs)
- Band: steps, heart rate
- Sleep window

Training & validation

- HUPA-UCM: 25 T1D, Spain Libre 2 + Fitbit (CC BY 4.0)
- T1D-UOM: UK, external test CGM + Garmin (CC BY 4.0)
- Synthetic Indian cohort (demo only)

2 PREPARE

Personalisation

- Fit twin on child's past nights
- Least squares, 5 parameters
- Insulin sensitivity anchored to the 1800 rule
- Refit as new nights arrive
- Assimilate today's readings

Ingestion & features

- Unified 5-minute timeline
- mg/dL, U, g; time-aligned
- Insulin on board (exp. model)
- Carbs on board
- Unlogged meals: 500 rule
- Glucose lags, slopes, SD
- Heart-rate deviation
- Events: <70 mg/dL for 15 min

3 MODELS

A · Physiology twin

- Bergman glucose–insulin model
- SC rapid + NPH / glargine depots
- Gut absorption, exercise effect
- 500-member night ensemble
- → P(low), time, depth
- What-if: snack, dose, NPH timing
- Counterfactual explanations

B · Night guard (LightGBM)

- Low within 60 / 120 min
- Glucose at +30 / 60 / 120 min
- 15-min smoothing, alert threshold
- Patient-held-out + external test
- 97% of night lows warned ≥15 min

C · Sensor priority

- Week-1 trial-sensor ranking
- Spearman 0.63 on 29 patients

4 SERVE

FastAPI service

- GET /api/board
- GET /api/patients/{id}
- POST …/whatif
- GET /api/validation
- Twin cache per child
- Serves dashboard
- One command: ./run.sh

5 USERS

Doctor dashboard

- Overview
- Tonight Board (9 PM)
- Twin view + why
- Rehearse a plan → approve
- Night guard replay
- Sensor Priority
- Evidence

Family channel

- WhatsApp / voice call
- Kannada, Hindi, Malayalam…
- Works on feature phones
- Only doctor-approved actions
- Night alert ~1 h before a low

Trust layer

DPDP Act 2023 s.9: verifiable parental consent · clinic is the data fiduciary (Rules 2025, Sch. IV health exemption) · no ads / analytics on children's data
Doctor-in-the-loop decision support · FHIR R4 / ABHA-ready · patient-held-out + external validation · honest limits published · Apache-2.0

One night with TwinPulse

What happens, when, and who sees it

All day	**Data arrives** CGM or finger-pricks, insulin doses, meals, steps and heart rate stream into the child's 5-minute timeline.
21:00	**Twin rehearses tonight** Each child's twin is refitted on past nights, assimilates today's readings, and lives 23:00–07:30 500 times.
21:00	**Tonight Board** The clinic sees every child ranked by P(low). The diabetes educator calls the red rows before bed.
21:15	**Doctor rehearses a fix** Bedtime snack, smaller dinner dose, or NPH at bedtime: the twin re-lives the night with each plan.
21:30	**Plan approved and sent** The family receives only the approved action, in their language, by WhatsApp or voice call.
23:00–07:00	**Night guard (CGM)** Every 5 minutes: P(low within 2 h). Above threshold → voice call to the parent, median 65 min before the low.
Morning	**Learn** The night's real data refits the twin. Over 14 days the trial sensor ranks who needs a CGM most.